

Ali Zamani

Senior ML/AI Engineer @ Loblaw Digital

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Relevant Work Experience

Senior ML/AI Engineer (Full-Time)

Jun. 2025 - Present

Loblaw Digital, Toronto, ON

- Designed a **planner-based agentic AI system** with specialized agents, pre/post-guardrail layers, and long-term, short-term, episodic, and pattern-based memory architectures.
- Built a production **hybrid-search Agentic RAG System** using Vertex AI, SPLADE, and Qdrant with Airflow-orchestrated Cloud Run indexing and blue/green promotion; delivered a business UI using K8S for document retrieval achieving 97% hit rate, 78% precision, 97% recall.
- Developed and integrated an **Agent Tool Package** (built with LangGraph) to support internal AI agents for customer-facing automation and orchestration of modular agent workflows.

Machine Learning Engineer (Full-Time) 🎥

Apr. 2023 - Jun. 2025

Priceline (Booking Holdings), Toronto, ON

- Built a **standardized ML pipeline** for preprocessing, training, and online/offline deployment on **Vertex AI**, featuring modular design, feature store integration, versioning, and backfilling. Orchestrated the pipeline with a reusable **Airflow DAG**.
- Designed a scalable **GenAI framework** on GCP with CI/CD integration, featuring a robust **Inference module** with rate limiting, monitoring, parallel processing, and use-case-specific configuration, and a **Validation module** using **LLM-as-a-Judge** for automated evaluation and continuous quality feedback. Leveraged **LangChain** for orchestrating dynamic LLM workflows and integrating retrieval-augmented generation capabilities.
- Delivered a **hotel review summarization** use case combining sentiment analysis and summarization, with an algorithm to surface high-impact reviews. Achieved a **30% conversion rate uplift** through **A/B testing**.
- Built a **flight price prediction** use case using **time series forecasting**. Deployed on **Vertex AI** with data from **BigQuery**, enabling data-driven pricing and integration with product features.
- Supported the **Penny Chatbot** using **Vector RAG**, **fine-tuned models**, and **AI agents** to enhance customer interactions through contextual understanding and dynamic response generation.

Data Scientist (Contract Position)

Sept. 2022 - Dec. 2022

AltaML, Calgary, AB

- Implemented a **LightGBM** and **XGBoost** algorithm for predicting the permeability of rock core images with an accuracy of **94%**, saving upwards of 10 million dollars for the client.

Data Scientist and Chatbot Developer (MIRA Chatbot - 🎥)

Jan. 2021 – Sept. 2022

Department of Computing Science, University of Alberta and Amii, Edmonton, AB

- Developed and deployed the back-end and front-end of the **MIRA chatbot** (mymira.ca), enhancing user experience with advanced NLP.
- Evaluated various Recurrent Neural Network language models for intent detection and entity extraction, achieving an F1-score of **97%** for intent detection and **83%** for entity extraction.

Education

M.Sc. in Computer Science

University of Alberta, under the supervision of *Dr. Osmar R. Zaiane*

Jan. 2021 – Aug. 2022

Edmonton, AB, Canada

M.Sc. in Computer Engineering

Amirkabir University of Technology

Sep. 2017 – Sep. 2020

Tehran, Iran

B.Sc. in Electrical Engineering

Kashan University

Sep. 2013 – Sep. 2017

Isfahan, Iran

Technical & Soft Skills

Programming: Python (5+ years), C++, C, JavaScript

Generative AI / Agents: LLMs, multimodal models, LangChain, prompt engineering, agent orchestration, tool integration

Cloud / Infra: GCP (Vertex AI, BigQuery, Cloud Storage), Azure, Docker, Kubernetes, Airflow

Libraries: TensorFlow, PyTorch, Hugging Face, Scikit-learn, NumPy, Pandas

Tools: Docker, Git, Spark, Flink, NGINX, Jira

Database: SQL, PostgreSQL, MySQL, BigQuery

Frameworks: Rasa, LangChain, Django, Flask

Software Skills: Communication, Stakeholder Collaboration, Leadership, Ownership, Agile Development

Selected Projects

Google Hackathon (Placed 1st and 3rd in Consecutive Years)

- **Neighborhood Recommendation System:** Leveraged vector embeddings to suggest neighborhoods to users based on their preferences, enhancing personalized travel experiences.
- **Travel Assistance Chatbot:** Developed a chatbot using GPT-4 to assist customers in finding their desired travel destinations, significantly improving user engagement and satisfaction.

ML-Pipeline Template 🔄

- Developed an ML pipeline template to create a user friendly utility to drastically speed up the development and implementation of a machine learning model for all sorts of various problems.

Microsoft Hackathon

- **Burnout Detection System:** Built an ML pipeline on Azure to detect burnout among call center agents, utilizing a pre-trained transformer-based model (BERT) for accurate sentiment analysis and stress detection.

Hotel Revenue Forecasting

- Forecasting global monthly hotel revenue for 2024 using time series and machine learning models.

Spotify Music Recommender 🔄

- Spotify Music Recommender is a machine learning project that leverages the Spotify dataset to build a personalized music recommendation system. The goal is to explore various recommendation techniques—including content-based filtering, clustering, and deep learning—to recommend tracks tailored to user preferences.

Selected Publications

Developing and Implementing a Mental Health Chatbot to Support Healthcare Workers

2021, A.Zamani, M.Gharayat, J.Nobel, O.Zaiane, E.Stroulia – REMAP-D, Vancouver, British Columbia, Canada